Quality-Focused Internet of Things Data Management: A Survey, Perspectives, Open Issues, and Challenges

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***Abstract*—The integrity of Internet of Things (IoT) devices has caused a fast spread in an era of data-driven decision-making across businesses. This tutorial survey provides a comprehensive review of current IoT data handling advances, focusing on data quality management (DQM). This article starts with the key aspects of IoT data management. In this regard, we shed light on the data source, volume and velocity, variety, lifecycle, security and privacy, scalability and distribution processing, anomaly detection, and energy efficiency. Then, we present a comprehensive taxonomy of IoT DQM based on the application type, such as smart cities, healthcare, agriculture, environmental monitoring, retail and supply chain, and smart grids (SGs). As IoT data processing, analysis, and security play a significant role in DQM, this tutorial survey carefully addresses how modern technologies maintain this role. More particularly, this work investigates the use of edge computing for real-time data processing and the incorporation of synthetic data to supplement restricted resources, incorporating the issues of managing the massive datasets created by IoT implementations. In addition, this article addresses the use of machine learning (ML) algorithms for in-depth analysis of IoT data streams, DQ evaluation protocols, and detection tactics during data transfer. Moreover, this article investigates complete security measures for protecting sensitive data, such as access control regulations and several security techniques, including authentication, encryption, and secure communication protocols that enable IoT data management. Besides, blockchain technology's significant roles in this regard have been comprehensively addressed. Along with summarizing and reviewing the latest efforts in DQM in IoT-based systems, we shed light on their strong and weak points and discuss upcoming trends and potential difficulties in IoT data management. Last but not least, we continue by emphasizing the cumulative impact of these advances and shedding light on the open issues and challenges. Finally, this in-depth tutorial survey aims to be a significant resource for academics, practitioners, and stakeholders interested in the changing environment of IoT data management, with a particular emphasis on DQ.**

***Index Terms*—Blockchain, data management, edge computing, Internet of Things (IoT), machine learning (ML), security, smart systems.**

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NOMENCLATURE

5G and 6G	Fifth generation and sixth generation.
AI	Artificial intelligence.
ANN	Artificial neural network.
AM-DR	Adaptive method for data reduction.
CNN	Convolutional neural network.
CoAP	Constrained application protocol.
CSMA-CA	Carrier sense multiple access with collision detection.
DEDaR	Distributed energy-efficient data reduction.
DL	Deep learning.
DQ	Data quality.
DQM	Data quality management.
DRLS	Desired reliability level scheme.
DWU-ODBN	Dual weight updation-based optimal deep belief network.
E2EE	End-to-end encryption.
GRU	Gated recurrent unit.
FL	Federated learning.
FTTH	Fiber-to-the-home.
GDPR	General Data Protection Regulation.
IDPS	Intrusion detection and prevention system.
IoT	Internet of Things.
IIoT	Industrial Internet of Things.
LGB	Light gradient boosting.
LoRaWAN	Long-range wide area network.
LPWAN	Low-power wide area network.
LSTM	Long short-term memory.
MAE	Mean absolute error.
MIMO	Multi-input–multi-output.
ML	Machine learning.
MQTT	Message queuing telemetry transport.
mse	Mean square error.

NB	Narrowband.
QoDID	Quality of data for IoT devices.
RBS	Reliability-based subchannel scheme.
RMSE	Root-mean-square error.
SBRS	Standby route selection scheme.
SVM	Support vector machine.
TDMA	Time-division multiple access.
UTM	Unified threat management.
WSN	Wireless sensor network.

I. Introduction

THE proliferation of IoT devices has catalyzed a transformative shift in how data are acquired, processed, and utilized across various domains [1]. This technological advancement has inaugurated an era of unparalleled connectivity, facilitating seamless interactions between the physical and digital realms [2]. Consequently, IoT ecosystems have evolved into indispensable components of enterprises spanning healthcare, transportation, smart cities, and industrial automation [3].

The effective management of the vast and dynamic datasets generated by IoT deployments is a critical prerequisite for their efficacy [4]. Indeed, the datasets are often characterized by some specs such as volume, velocity, and variety. Besides, the datasets necessitate complex storage, processing, and analytical methodologies [5]. Accordingly, it is strictly desired to guarantee the security and integrity of sensitive data within these networked systems. Li et al. [6] presented a perspective of the current advancements in IoT data management, encompassing a wide spectrum of pivotal domains along with security threats.

IoT systems are also engaged with several modern technologies, such as edge computing, which represents a pivotal innovation in tandem with big data [7]. This integration relies on a decentralized connection, which is beneficial for the computational resources. It also enables real-time processing at the data source and reduces the substantial latency and bandwidth requirements [4]. Moreover, data synthesis-based generative models emerge as a potential strategy for enhancing constrained datasets, opening the way for more precise and comprehensive analysis [5].

The rapid advancement of ML has transformed numerous domains by enabling automation, deriving meaningful insights from vast and complex datasets, and supporting intelligent decision-making [8], [9], [10], [11], [12]. Today, ML models are widely integrated into diverse applications, including image recognition, natural language processing, autonomous systems, and personalized services. As these systems grow in complexity and computational demands, there is a need to enhance their efficiency, adaptability, and reliability [13], [14], [15], [16]. In the context of IoT DQM, these challenges become even more pronounced. High-quality, reliable data are essential for training and deploying robust ML models in IoT environments. Poor DQ, e.g., missing, noisy, or inconsistent data, can compromise model performance and lead to inaccurate or biased outcomes. Thus, improving ML system performance is not only a computational concern but also a fundamental requirement for ensuring accurate, trustworthy, and scalable data-driven solutions in IoT ecosystems [17], [18], [19]. ML also plays a significant role in extracting valuable features from IoT datasets [20]. The deployment of ML techniques can handle the unique challenges presented by IoT contexts [21]. It also addresses vital concerns such as quality assessment and anomaly detection, ensuring reliability and integrity [22], [23].

Without loss of generality, data transmission prioritization in IoT systems is desired to satisfy the resource constraints for energy management [24]. Accordingly, developing efficient optimization approaches for data transmission while upholding data integrity and conserving resources is carefully considered [25]. Furthermore, robust security techniques, such as access control policies and blockchain technology, are periodically investigated and developed against unauthorized access and data breaches [26], [27], [28], [29], [30].

In the rapidly evolving applications employing the IoT, managing the different data formats generated by interconnected devices presents a significant challenge. More particularly, quality-focused IoT data management has captured the focus as a critical area of research. It ensures valuable aspects across diverse applications, such as reliability, accuracy, and timeliness of data. The IoT refers to a swiftly expanding network of networked devices and sensors that gather and exchange data, facilitating a wide range of applications and services [31]. The effective management of the substantial quantity of data produced by devices poses a main difficulty within this field [32].

While numerous studies have addressed specific aspects of the intersection of DQM practices within IoT ecosystems, e.g., big data processing, edge computing, and ML-driven analytics—there remains a pressing need for an integrated approach that prioritizes DQ.

Indeed, the emerging use of IoT networks, including a span of data types, requires the big data analysis role to efficient DQM [33], [34]. The gadgets in question, which encompass sensors and smart devices, generate a substantial amount of data, which is notable for their vast quantity, rapid production rate, and many formats [35], [36]. Undoubtedly, the professional analysis of big data will meet the advancement of intelligent IoT applications and contribute to the efficient extraction of data insights [37], [38]. In other words, such efficient analysis can contribute to capturing, memorizing, and appropriately processing the data [39], [40]. These factors are of utmost importance in effectively managing the substantial data flow [41], [42]. The examination of large-scale data for the IoT is an area of research that is inevitably changing, requiring ongoing creativity and the creation of new methods and solutions to address the accompanying difficulties and effectively utilize the vast capabilities of IoT data [43].

Unlike previous studies that discuss data collection, storage, or processing techniques in isolation, this article comprehensively examines optimizing these aspects to maintain high DQ across various IoT applications. Besides, this article explores the state-of-the-art paradigms and frameworks developed to address the complexities of IoT data management, with a particular emphasis on quality assurance. Moreover, by highlighting recent advancements, providing a critical analysis of quality-focused approaches, pinpointing the security measures, and conducting data analysis and processing.

This tutorial survey aims to fill the gap between the above aspects by providing a comprehensive study that not only reviews existing solutions but also highlights emerging trends and perspectives that bridge the gap between data management practices and the growing demand for reliable, driving efficient, anomaly detection, security fronts, intelligent systems, and high-quality IoT data through advanced technologies like blockchain and cloud systems focusing on the period from

TABLE I
COMPARISON WITH THE MAIN RELATED CONTRIBUTIONS IN THE LITERATURE

Ref	Scope	Technologies Covered	Applications	Methodologies	Unique Contributions	Open Challenges
[44]	6G and IoT convergence	Edge intelligence, Terahertz, space-air-ground	Healthcare, Vehicular network, UAVs, Satellite, Industry	Holistic review of 6G technologies for IoT	Comprehensive analysis of 6G roles in specific IoT domains	Research challenges in 6G-IoT integration and scalability
[45]	IoT in 5G wireless systems	5G NR, MIMO, mmWave, LPWANs, AR in IoT	Smart services, and Low-power networks	Layered analysis of 5G IoT system architecture	In-depth coverage of AR and low-latency requirements in 5G IoT	Gaps in 5G IoT security, real-time anomaly detection
[46]	IoT management technologies	RESTCONF, CoAP, LwM2M, SNMP	Device management	Comparative protocol analysis, taxonomy	Detailed comparison of traditional vs modern IoT management protocols	Challenges in standardizing and optimizing IoT management
[47]	Intelligent IoT	Artificial intelligence (AI), ML, smart networks	Healthcare, Smart cities, Vehicular Network, Industry	Analysis of security and privacy in IIoT	Focus on AI-driven security countermeasures in IIoT	Data privacy and confidentiality challenges in IIoT networks
[48]	Real-time IoT analytics	Big data, real-time processing	Industry, Smart cities	Review of analytics network methodologies	Focus on IoT analytics network inadequacies	Research challenges in optimizing analytics networks for real-time data
[49]	Resource allocation in 5G HetNets	HetNets, RA models, learning-based RA	HetNets	Classification of RA algorithms for HetNets	Proposes structures for RA in 6G HetNets	Challenges in mutual interference and spectrum sharing
[50]	IoT data analytics with deep learning	Deep learning, IoT streaming data, big data	Smart devices, and Fog/Cloud	ML/DL techniques for IoT data analytics	Comprehensive coverage of DL techniques for IoT data	Potential challenges in real-time DL-based analytics in IoT
[51]	Smart city IoT infrastructure	Wireless networks, AI, big data	Sustainability and Resource management	Dynamic task scheduler for Apache Spark	Improved task scheduling for IoT big data in smart cities	Sustainability and privacy challenges in urban IoT infrastructure
[52]	IoT data lifecycle management in smart cities	Data lifecycle management, security, privacy	Smart city	Data management lifecycle approach	Data-centric view of smart city IoT management	Privacy, consistency, and interoperability challenges
[53]	Unified threat management (UTM) for home networks	Firewall, intrusion detection, antibot protection	Home network security	Integration of UTM components	A survey on integrated UTM systems	Limited computational resources for UTM in home networks
[54]	Security of 5G-enabled IoT	5G, IoT, security frameworks	5G applications	Review of security frameworks in 5G-IoT	Proposes modern open-source standardization	Security and privacy concerns in 5G-IoT device integration
This paper	IoT data handling, DQM, data privacy, security, processing, and analysis	Edge computing, ML, Blockchain, DQM IoT protocols, IoT topologies, and smartness	Smart cities, Healthcare, Agriculture, Industry, Environmental monitoring, Smart homes, Edge computing, Big data, Retail and Supply Chain, and Smart grids (SGs)	Comprehensive taxonomy, anomaly detection, and real-time data processing and analysis	Focus on IoT DQM, synthetic data, energy efficiency in IoT systems, modern technologies for IoT data variety processing, analysis and security, and security protocols	AI-powered data insights, edge computing maturity, Interoperability and standardization, security in the age of pervasive connectivity, scalability and resource optimization, ethical considerations and data governance, environmental sustainability, energy efficiency, and data analysis and processing

2016 to 2025. Table I shows a holistic comparison between the main contributions in the related state of the art and the current tutorial survey. To the best of our knowledge, no similar tutorial has comprehensively considered the versatility of perspectives, open issues, and challenges for the quality-focused IoT management. Nomenclature presents the utilized abbreviations throughout this tutorial survey.

In this tutorial survey article, we delve into various critical aspects of IoT data management, aiming to provide a comprehensive understanding of the field. This article's involved topics are depicted in Fig. 1. Aside from summarizing the main related contributions presented in the literature, our contributions are outlined as follows.

1) Offering an interdisciplinary and end-to-end overview of IoT DQM, unlike prior works that tend to focus narrowly on one or two aspects (e.g., edge computing or security).
2) Combining perspectives from big data applications, ML, blockchain, cloud computing, and energy-efficient communication.
3) Presenting a comparative analysis of state-of-the-art literature, clearly outlining how our work fills gaps left by previous surveys. To the best of our knowledge, no previous tutorial has provided such a detailed and structured comparison across these many facets of IoT DQ.
4) Introducing a novel taxonomy that not only classifies current technologies and methods but also emphasizes emerging trends such as anomaly detection, real-time processing, and intelligent data-driven systems, thereby offering readers a forward-looking view of the field.
5) Uniquely integrating security and anomaly detection considerations into the discussion of DQ, an area often underexplored in other surveys but crucial for IoT environments where data reliability directly impacts system safety and decision-making.
6) Reviewing solutions and highlighting open research challenges and areas for future exploration, which we believe is essential for guiding new research efforts in this evolving domain.

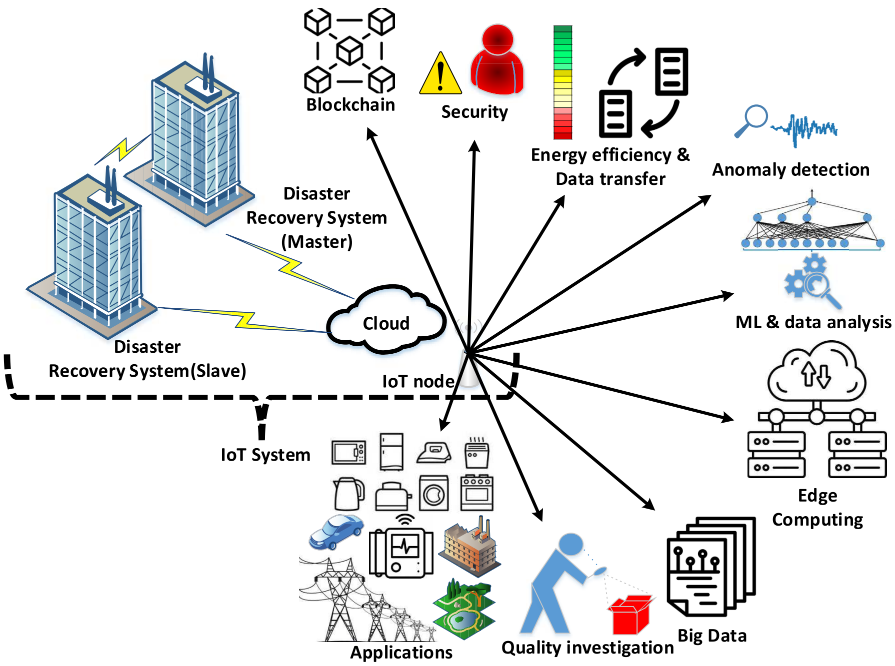

Fig. 1. Involved topics in this tutorial survey.

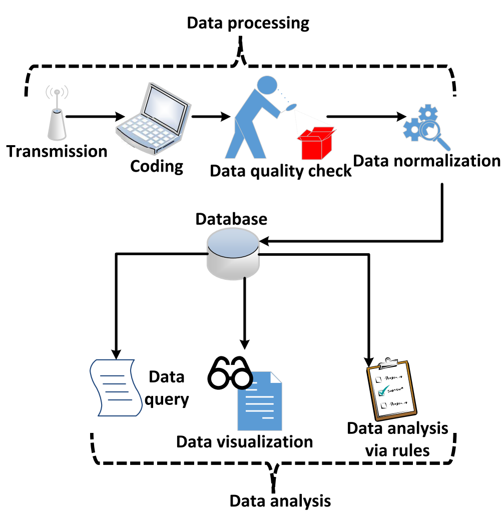

Fig. 2. Architecture of data management.

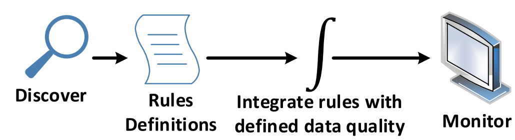

Fig. 3. General architecture of in IoT DQM.

This article is organized as follows. Section II indicates the key aspects of managing the IoT data. Section III presents a taxonomy of IoT DQM relying on the application type. Then, the roles of modern technologies evolved for IoT DQM in processing capabilities, analysis, and security fronts have been portrayed in Section IV. After that, the tutorial survey sheds light on the open research issues and future direction in Section V. Finally, this article is concluded in Section VI.

II. Key Aspects of IoT Data Management

Recently, IoT nodes have been deployed in many applications due to their capabilities such as reconfigurability, suitable size, environment versatility, ease of management, and diversity in technology. This section addresses the main key aspects of DQM in IoT systems. Fig. 2 outlines a comprehensive architecture for general data management, where the process starts with data transmission and coding, followed by quality checks and normalization to ensure data integrity. Once standardized, the data are stored in a database, supporting operations such as data querying, rule-based analysis, and data processing. Analytical insights are then derived through data analysis and effectively communicated via data visualization, completing the cycle from raw data acquisition to decision support.

DQ plays a crucial role in IoT data management, which ensures the reliability and accuracy of information used in decision-making, processing, and engineering systems [55]. Fig. 3 illustrates a systematic IoT DQM architecture, emphasizing a rules-driven approach. It begins with the definition and discovery of DQ rules, which are then integrated into monitoring systems. These mechanisms ensure continuous evaluation of data streams, enabling real-time detection of inconsistencies and maintaining high data reliability throughout the IoT ecosystem. Due to the diverse nature of data generated by IoT devices, it presents challenges that can significantly impact the effectiveness of the used applications [56]. As a result, several IoT-based systems need to enhance their DQM systems to fully integrate the IoT features and realize substantial benefits [57]. The importance of DQ in the IoT context is well-recognized by both experts and academics, highlighting the need for frameworks and methods to develop, evaluate, and improve DQ in IoT systems [58].

Kim et al. [59] proposed an approach to improve DQM processes in the IoT. That model included a process reference model (PRM) for IoT DQM and a recognition model called IoT DQM3. Another study has shed light on the IoTDQM

TABLE II
QUALITY IN IoT DATA MANAGEMENT

Ref	Description
[55]	Emphasizes the importance of DQ in IoT data management for reliable decision-making and system effectiveness.
[56]	Highlights concerns about the heterogeneous nature of data from IoT devices and its impact on subsequent applications.
[57]	Stresses the need for enterprises to enhance their DQM systems for effective IoT integration and benefits.
[58]	Acknowledges the widespread recognition of DQ's importance in IoT, emphasizing the need for frameworks to develop, assess, and improve it.
[59]	Introduces a framework for improving DQM processes in the IoT, including a PRM and a maturity model (IoT DQM3).
[60]	Presents a method for handling DQ in contexts involving 'smart, connected products' via international standards like ISO/IEC 25000 and ISO 8000.
[61]	Proposes a comprehensive framework for assessing DQ in the IoT using a knowledge graph-based architecture, illustrated with a practical use case.
[62]	Introduces a model-driven architecture-based approach for DQM in the IoT, addressing diverse needs and the subjective nature of DQ.
[63]	Introduces the QoDID framework to improve IoT device and sensor data collection and processing.

validation [60]. That work introduced a methodology for effectively handling DQ along with intelligent connected products. More particularly, that technique employed internationally recognized standards such as ISO/IEC 25000 and ISO 8000, which offered a collection of optimal strategies for evaluating and enhancing the DQ. Similarly, Khokhlov and Reznik [61] presented a comprehensive approach for IoT DQ assessment. In that work, using a knowledge graph-based architecture can facilitate the integration of DQ assessment into a multitude of IoT applications. Accordingly, such a tool can enhance the identification and integration of DQ measures. Moreover, the effectiveness of that work appears in addressing practical tasks. Another research effort has been developed via a model-driven architecture for IoT DQM [62]. It addressed the subjective nature of DQ and the multiuse of data consumers by allowing developers to express DQ requirements through models and a graphical editor. That approach automatically generated a customized infrastructure for DQM, tailored to the data consumer's specifications. Its flexibility and efficiency were demonstrated through the generation of two DQM infrastructures tested in a real-life IoT data stream scenario.

Al-Masri and Bai [63] introduced the QoDID framework to address the IoT system's difficulty in gathering and processing raw sensor data, which describes QoDID architecture, use cases, and implementation. Additionally, they show how the QoDID architecture may improve IoT device and sensor data collection, delivering clear insights. Finally, the main aspects of DQ are summarized in Table II. Here, we also present a comprehensive taxonomy for IoT nodes considering environment type, network topology, sensor type, used protocol, standard, and sensor specifications as depicted in Table III. In the following, we shed light on other significant IoT data aspects such as sources, volume and velocity, variety, lifecycle, security and privacy, scalability, and distribution.

A. Data Sources

Data are derived from a diverse array of sources, encompassing sensors that are integrated into a multitude of products, such as smart appliances, industrial equipment, environmental sensors, and wearable gadgets [75]. The sensors consistently gather data on several characteristics, including temperature, humidity, location, and additional factors [76]. In the context of a smart home environment, data sources encompass various devices such as thermostats equipped with temperature sensors, security cameras integrated with motion sensors, and wearable devices featuring heart rate monitors [75]. In the realm of industry, the collection of data can be derived from several sources, including sensors integrated into production machinery, energy meters, and GPS trackers affixed to transportation vehicles used for deliveries. A comprehensive comprehension of the various categories and variations of data sources inside is crucial in the process of devising data management strategies tailored to IoT applications. Every sensor produces data that possess distinct features. For instance, temperature sensors provide continuous numeric data, whereas image sensors produce large volumes of image data [76]. The identification of these distinctions facilitates the selection of suitable data storage formats and processing techniques [77]. Another essential point to shed light on the heterogeneity of IoT data (e.g., continuous sensor data versus multimedia data) introduces distinct vulnerabilities and quality issues, such as noise, data inconsistency, and attack surfaces specific to each source type [78], [79].

B. Data Volume and Velocity

One of the distinguishing features of data is its considerable magnitude and rapidity [80]. Devices produce substantial volumes of data at rapid intervals, posing challenges for conventional data management solutions [81]. The management of this influx of data necessitates the utilization of specific procedures and infrastructure. In the context of a hypothetical situation, envision a smart urban environment wherein the traffic management system effectively acquires data from plenty of vehicle sensors, traffic cameras, and GPS devices promptly. The volume of data generated has the potential to rapidly accumulate to the scale of terabytes or perhaps petabytes within a relatively short time frame. The management of large quantities of data involves the utilization of distributed storage systems and parallel processing frameworks [82]. The pace of data is equally crucial. Certain applications necessitate immediate and timely replies. Autonomous vehicles, for example, depend on real-time data processing in order to make informed driving judgments. To effectively manage IoT data, it is imperative that data management systems possess the ability to analyze data streams with minimal latency. The considerations of data volume and velocity in the context have a significant influence on the selection of storage systems, data compression methods, and processing pipelines for the management of IoT data [83]. It is worth elaborating how the massive influx of data strains storage, processing pipelines, and real-time analytics, thereby heightening risks of data loss, corruption, and delayed anomaly detection. Therefore, data characteristics, quality assurance, and security management in IoT ecosystems can be enhanced [84], [85].

C. Data Variety

The data encompass several types, such as structured data (e.g., sensor readings), semistructured data (e.g., JSON or

TABLE III
TAXONOMY FOR KEY ASPECTS OF IoT DQM

Application	In or Out-door	Topology	Sensor type	Manufacturer	Protocol	Standard	MAC	Advantage	Specifications (Coding, Chip, Temperature)	Challenge
Robot Control & Health monitoring [64]	Both	Star, Mesh [65]	Tmote Sky	University of California, Berkeley	Zigbee [64], 6LoW-PAN [65]	IEEE 802.15.4	CSMA and TDMA	Supports external flash	TinyOS & CC 2420 & -40°C- 85°C	High hardware preparation
Health monitoring, ECG, PIR, Galvanic, Skin Response [66]	Both	Tree [66]	SHIMMER	Intel	Bluetooth Protocol	IEEE 802.15.1	CSMA and TDMA	Supports Bluetooth and micro SD card	TinyOS & CC 2420 & -40°C - 80°C	Form factors shortage
Smart cities [65]	Both	Star and IoT paper	Mica2	CrossboW	No predefined protocol	3G	CSMA-CA	Supports ad-hoc mesh networking	TinyOS & CC 1000 & -40°C - 85°C	High hardware preparation
Bluetooth [67]	In	All topolo-gies	EZ430-RF2480-F2500T	Texas Instrument	Zigbee	2.4GHz ZigBee	CSMA-CA	Supports USB interface and high data rate	C & CC 2480 & -40°C - 85°C	Idle and sleep modes consume quite high power
Wind, temperature, soil, light, humidity, moisture [68]	Both	Mesh [67]	IRIS	Crossbow	Token ring [69]	IEEE 802.15.1 and ISM	CSMA-CA	Supports high data rate	TinyOS & Atmel & -40°C - 85°C	Compatibility
Solarium application, space and magnetic activities, robotics, art, toys, electronics [70], [71]	Both	Large scale magnetic topology [70], [71]	SunSPOT	Sun Microsys-tems	Zigbee and LowPAN	2G and 6G, IEEE 802.15.4	CSMA-CA	SW and HW are open source	J2ME & CC 2420 & -40°C - 85°C	Costly
Smart cities, autonomous computing, eHealth [65]	Both	Array topology [72]	Micaz	CrossboW	No predefined protocol	IEEE 802.15.4 and ISM band	CSMA-CA	Supports ad-hoc mesh networking	TinyOS & CC 1000 & -40°C - 85°C	High hardware preparation
LoraWAN [73], [74]	Both	Star, tree, mesh	Wasp-motes	Libelium	LoRaWAN	IEEE 802.15.4	CSMA-CA	Upgrade or change of firmware versions without physical access	C++ & CC 2500 & -40°C - 70°C	Early switch off before critical low-power; Outdoor operation much less indoor

XML files), and unstructured data (e.g., photographs or text from IoT devices). The handling of this variability requires the implementation of adaptable data management solutions [46], [86], [87]. Efficiently processing and integrating these different data formats is essential for gaining insights from the entire dataset. To effectively handle the diversity of data, data preparation pipelines are frequently used [88]. The primary function of these pipelines is to perform data cleansing and transformation operations, resulting in the conversion of data into a standardized format. This process can facilitate the subsequent analysis and contribute to data integration with preexisting databases or analytics tools.

D. Big Data and Large Language Models

The integration of big data technologies and large language models (LLMs) is emerging as a powerful paradigm in IoT DQM, addressing challenges of scalability, real-time analytics, and data interpretation. Big data frameworks such as Apache Hadoop, Spark, and Flink enable efficient handling of the massive, high-velocity data streams generated by IoT devices through distributed storage and parallel processing [89]. These capabilities are essential for applications that require continuous, low-latency insights, such as smart cities, industrial automation, and digital healthcare. The LLMs can bring semantic intelligence to IoT systems, allowing for natural language querying, contextual anomaly detection, and meaningful summarization of unstructured sensor data [90]. Their ability to perform cross-modal reasoning across textual, visual, and numerical inputs enhances the decision-making capabilities of complex IoT environments [91]. This convergence of big data infrastructure with LLM-powered intelligence is reshaping the future of IoT systems, providing the way for more adaptive, interpretable, and accessible data management frameworks, especially as research advances toward lightweight, edge-compatible, and privacy-preserving AI models.

E. Data Lifecycle

The lifecycle of IoT data involves several stages, including data collection, storage, processing, analysis, and eventually, archiving or destruction [92]. Effective management for each stage of this lifecycle is crucial for efficient data management. In the initial stage, data are collected from devices and stored in its raw, unaltered form. Then, the data often undergo preprocessing steps, e.g., cleaning, filtering, and transforming [93]. This stage is used to organize the data into a more structured format for storage. Afterward, the data become available for a variety of uses, such as real-time analysis, historical trend tracking, or ML tasks [50]. Over time, as the data become less relevant, data retention policies are utilized, leading to either archiving or deletion. In fact, proper management of the data lifecycle requires careful consideration of factors such as storage costs, access speed, and compliance with data governance regulations [94]. For instance, legal requirements may dictate how long certain types of data, such as health data from wearable devices, must be kept. To stay compliant and optimize resources, organizations must adopt well-defined data lifecycle management strategies [95].

F. Data Security and Privacy

Due to the rise of connected devices, the need to ensure the security and privacy of IoT data has become desired. Unauthorized access to sensitive information can have severe consequences, making it crucial to implement strong security measures such as encryption during transmission and storage, authentication, and intrusion detection systems [96], [97]. Privacy-preserving techniques, such as data anonymization and differential privacy, are also vital to protect personal data [98]. The interconnected nature of IoT systems complicates the task of securing data as it flows from devices to cloud or edge nodes, requiring continuous monitoring and updates. In smart healthcare, for example, encrypting data from wearable devices and restricting access to authorized personnel is essential to protect patient privacy [99]. Moreover, compliance with data protection regulations, e.g., GDPR, is critical in managing data to avoid privacy violations, especially as IoT systems can aggregate data into detailed profiles of individuals' behaviors [100]. To provide a broader and more global perspective on data security in IoT environments, we compare major regulatory frameworks, including the EU's GDPR, the U.S.'s CCPA, the Health Insurance Portability and Accountability Act (HIPAA) [101], and China's Personal Information Protection Law (PIPL) [102]. While GDPR and PIPL impose strict consent requirements and control over cross-border data transfers, CCPA offers more flexibility with an emphasis on the right to opt out rather than opt in. HIPAA, on the other hand, focuses specifically on the healthcare sector. These variations significantly impact IoT data management strategies, particularly in terms of compliance, user rights, and data localization policies [103]. A comparison of key global data security standard impacting is presented in Table IV.

G. Scalability and Distributed Processing

To efficiently manage the growing number of devices and the massive data streams they generate, data management systems must be scalable. Accordingly, distributed processing and storage architectures have emerged, which provide both scalability and fault tolerance [82]. Scalability typically includes horizontal scaling, where additional servers or nodes are added to handle increased data loads. Key components of scalable IoT data management systems include distributed databases, stream processing frameworks, and load-balancing techniques [51]. Moreover, distributed processing not only supports scalability but also allows for real-time analysis of IoT data, ensuring timely insights [104]. Parallel processing

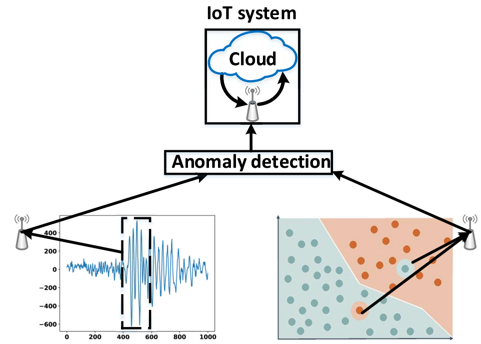

Fig. 4. Detecting data anomaly via an IoT system.

TABLE IV
COMPARISON OF KEY GLOBAL DATA SECURITY STANDARDS IMPACTING IoT DQM

Standard	Scope	Consent Requirement	Cross-Border Transfer	Sector Focus
GDPR	Personal data	Strong (opt-in)	Strict control	Technology
CCPA	Consumer data	Moderate (opt-out)	More flexible	Economic
HIPAA	Health data	Strong (opt-in)	Healthcare sector only	Healthcare Industry
PIPL	Personal data	Very strong (opt-in & localization)	Very strict	General

of data across numerous nodes enables expedited analysis and prompt reactions to dynamic circumstances. The establishment of dependability and fault tolerance in distributed systems holds significant importance in the context of IoT data management, as it mitigates the risks of data loss and system downtime that may arise from hardware failures or network disruptions [105]. An illustrative case is the implementation of a smart grid (SG) system responsible for the management of electricity distribution. During moments of high demand, the system is required to effectively manage an influx of data from smart meters and subsequently adapt power distribution in response [106]. An architecture that is both scalable and fault-tolerant is essential for maintaining uninterrupted service.

H. Anomaly Detection in IoT Data Transfer

The detection of anomalies is of utmost importance in guaranteeing the security and dependability of IoT data transfer, such as data stream or data category, as shown in Fig. 4. An anomaly can be defined as a deviation or irregularity observed in the data generated by the IoT, which does not conform to the expected patterns or behaviors [107]. Anomalies may manifest as a result of diverse circumstances, encompassing sensor faults, cyberattacks, system failures, alterations in the environment, or even human errors. The identification of anomalies holds significant importance in ensuring the preservation of the integrity, security, and performance of IoT networks and applications [108]. The timely identification and response to anomalies can effectively neutralize possible threats or operational concerns, thereby minimizing harmful consequences such as data breaches, equipment damage, or service outages.

IoT anomaly detection solutions utilize sophisticated data analytics, ML algorithms, and statistical modeling methodologies to discern patterns and anomalies within the data [23]. The methods discussed can be categorized into three main approaches: supervised (SV), unsupervised (UV), and semisupervised (SSV) [109]. SV approaches are dependent on the availability of labeled training data for constructing

TABLE V
ANOMALY DETECTION TECHNIQUES IN IoT DATA TRANSFER

Ref	Description	ML Type
SV	CNN-based intrusion detection model for IoT networks achieving high accuracy and F1 scores compared to existing approaches.	[111]
	Anomaly detection in IIoT data streams using a supervised contrastive spatiotemporal variational autoencoder with graph attention and adaptive model updates.	[112]
UV	Fog-empowered anomaly detection for time-critical IoT applications via Fog computing architecture and hyperellipsoidal clustering.	[113]
	FL-based anomaly detection in IIoT using CNN-LSTM model with real-time and lightweight detection.	[114]
	FL and CNN-LSTM framework for accurate and efficient anomaly detection in IIoT, surpassing traditional FL frameworks.	[115]
	Anomaly detection in WSNs using autoencoder for distributed detection without extensive resources.	[116]
	Anomaly detection model via the Hurst exponent vector and multifractal spectrum for identifying deviations in network traffic properties in IoT communication networks.	[117]
SSV	A hierarchical stacking temporal convolutional network for anomaly detection in IoT communication.	[118]
	DWU-ODBN: A framework for attack detection in IoT devices using preprocessing, feature extraction, feature selection, and classification.	[119]

models capable of discerning between normal and anomalous events. In contrast, UV algorithms are designed to detect anomalies without any prior knowledge of normal behavior, rendering them well-suited for identifying novel or unfamiliar patterns [110]. SSV methodologies include aspects of both SV and UV procedures, utilizing both labeled and unlabeled data to improve the precision of anomaly identification. Table V summarizes the main contributions of the ongoing evolution of the IoT development of novel anomaly detection approaches. More detailed discussions about the research efforts in this regard are in Sections II-H1–II-H3.

1) SV Models: Ullah and Mahmoud [111] proposed a novel anomaly-based intrusion detection model for IoT networks using CNNs. By leveraging the power of DL, particularly CNNs, the model achieves high accuracy, precision, recall, and $F1$-scores in detecting anomalies compared to existing approaches. The CNN-based multiclass classification model is validated using various intrusion detection datasets, showcasing its effectiveness in accurately identifying anomalies in IoT networks. Another effort has utilized a spatiotemporal variational autoencoder enhanced with SV contrastive learning, graph attention, and adaptive model updating for anomaly detection in IIoT [112].

2) UV Models: Lyu et al. [113] proposed a novel anomaly detection method called fog-empowered anomaly detection for time-critical IoT applications. By leveraging the fog computing architecture and an efficient hyperellipsoidal clustering algorithm, the method achieves accurate and timely detection of abnormal patterns. Unlike centralized and distributed methods, the end nodes in the fog computing architecture do not perform processing or clustering, leaving these tasks to the fog and cloud layers. Another work presented a framework for precise anomaly detection in IIoT using an FL approach [114]. It introduces a CNN-LSTM model that captures detailed features while utilizing LSTM units to predict time series data. The framework enables real-time, lightweight anomaly detection by incorporating a gradient compression mechanism. That model reduced communication costs by about 50% compared to traditional methods. Liu et al. [115] introduced an innovative anomaly detection framework for IIoT systems. It combined on-device FL with an attention mechanism-based CNN-LSTM model to build a robust anomaly detection model across decentralized edge devices while reducing communication overhead, outperforming traditional FL approaches.

Another research effort has been presented in [116], which proposed an innovative method for anomaly detection in WSNs using autoencoder neural networks. Their algorithm facilitates distributed anomaly detection directly at the sensors, minimizing the need for extensive communication or computational resources. The experimental results demonstrate high detection accuracy, low false alarm rate, and adaptability to changing environments. In the same way, Dymora and Mazurek [117] proposed another model for anomaly detection in IoT communication networks. Their model employed the Hurst exponent vector and multifractal spectrum to effectively detect irregularities in network traffic, which may capture malicious activity. By overcoming the limitations of existing intrusion detection systems, this approach provided a more robust solution for enhancing the security and reliability of IoT infrastructure.

3) SSV Models: Cheng et al. [118] introduced an SSV hierarchical stacking temporal convolutional network designed for anomaly detection in IoT communication. This model efficiently handles both labeled and unlabeled data, accounts for the unique characteristics of streaming data, and enhances the detection of anomalies with improved performance and efficiency. Besides, this method contributes to the advancement of anomaly detection in IoT networks by addressing the challenge of limited labeled data. Similarly, Sathya et al. [119] presented a framework for detecting attacks on IoT devices. Their approach incorporates preprocessing, feature extraction, feature selection, and classification to pinpoint the sources of attacks. The experimental results demonstrate that their method outperforms existing techniques, achieving an attack detection time of 77 s with an accuracy of 98.1%.

I. Energy-Efficient Data Transfer in IoT

The issue of energy efficiency in data communication within the context of the IoT is of the highest importance, given the widespread implementation of IoT devices and their frequently restricted energy resources [120]. Accordingly, observing the IoT nodes' energy consumption level is desired. The optimization of data transfer mechanisms has the potential to exert a substantial influence on the overall energy consumption of IoT systems, hence extending the operational lifespan of battery-powered devices. Numerous methodologies have been examined to get enhanced energy efficiency within the field of data transfer in the IoT [121], [122], [123]. One potential strategy is the utilization of data aggregation, which entails the consolidation of data obtained from several IoT devices into a unified packet [124]. This approach serves to decrease the frequency of transfers and mitigate energy usage.

Aggregation can be executed at different tiers, such as inside a localized cluster or at a centralized gateway, contingent upon the particular IoT framework.

A further approach involves the utilization of effective communication protocols. Traditional communication protocols such as MQTT and CoAP are extensively employed in IoT applications. However, it is worth noting that these protocols may not possess the highest level of energy efficiency [125]. Novel protocols, such as LPWAN technologies [126] like NB-IoT [127] and LoRaWAN [128], have been developed with the primary objective of addressing the energy limitations of IoT devices. These protocols provide an extended coverage range and reduced power consumption, making them suitable for energy-constrained IoT devices.

Indeed, power management strategies play a crucial role in improving the energy efficiency of data communication in IoT systems. More particularly, IoT devices can implement techniques such as duty cycling, where they switch between active and sleep modes to conserve energy during periods of inactivity [129]. In the literature context, an adaptive transmission power control model adjusted the transmission power based on the distance between devices to optimize the energy consumption [130]. Additionally, edge computing can be utilized to process data locally, reducing the amount of information that needs to be sent to the cloud and further enhancing energy efficiency.

Energy savings can be achieved by minimizing needless data transmission and cloud communication through the processing and analysis of data at the network edge, in analogy to IoT devices [131]. ML methodologies can also make a valuable contribution toward enhancing the energy efficiency of communication in the IoT. Through the utilization of local data patterns or event prediction, IoT devices can selectively communicate data, thereby diminishing the overall volume of transmitted data and effectively preserving energy resources [132].

Zhang et al. [133] proposed a cell-free IoT system based on a cell-free wireless communication network to enhance the reliability and sustainability of the IoT. By diluting the traditional cell concept, the cell-free IoT architecture improves system robustness compared to cellular IoT. This article addresses resource allocation and data transmission and presents an optimization model and a novel heuristic algorithm based on ML techniques. Through extensive simulations, the proposed cell-free IoT system demonstrates improved energy efficiency and achieves better reliability and sustainability compared to existing algorithms. Fathy and Barnaghi [134] introduced an AM-DR in IoT networks to minimize communication costs and energy consumption. Through AM-DR, data transmission between sensor nodes is significantly reduced while maintaining data accuracy. Real-world evaluations demonstrate up to 95% communication reduction without compromising prediction accuracy. The proposed approach, integrated with the LEACH protocol, effectively reduces energy consumption and prolongs the network lifetime, offering a promising solution for energy-efficient IoT applications.

Hussein et al. [135] proposed a DEDaR approach for IoT networks to minimize data transmission, storage, and energy consumption. More concretely, this model utilized autoregressive prediction to predict future data and make decisions on whether to transmit current data to the gateway. Moreover, the model eliminated the redundant data using an efficient compression method based on adaptive piecewise constant approximation, symbolic aggregate approximation, and fixed code dictionary with Huffman encoding. The model relied on real-sensed data by which the obtained simulation results demonstrated that the DEDaR approach exceeded the data reduction percentage of recent methods, transmitted more data size, preserved more energy consumption, and enhanced data accuracy. Finally, this approach provided an effective solution for optimizing data transmission and resource utilization in IoT networks. The exerted efforts for IoT energy efficiency have continued to work on Wi-Fi infrastructure to avoid monthly service charges and reduce power consumption [136]. In that work, an asymmetric physical design was employed to enable a broadband access point (AP) to communicate with multiple NB-IoT devices at a low sampling rate. In this regard, each IoT device tunes its carrier frequency to a specific subcarrier of the AP's signal, allowing efficient data encoding/decoding. The experimental results demonstrated that the model achieved high communication performance with 24 IoT devices, offering cost-effective and energy-efficient IoT connectivity.

Al-Kadhim and Al-Raweshidy [137] addressed the challenges of reliability and energy efficiency in the IoT by four optimization scenarios using a mixed-integer linear programming model. The scenarios include an SBRS for node failure recovery, a DRLS to minimize traffic power consumption while meeting reliability requirements, an RBS to mitigate interference, and a reliability-based data compression scheme (RBDS) to overcome link capacity limitations. The proposed schemes effectively balance reliability and power consumption, with average power savings of 57% in SBRS and 60% in RBDS compared to DRLS. These optimization approaches contribute to achieving reliable and energy-efficient data transport in critical IoT applications such as smart cities.

Stojkoska and Nikolovski [138] focused on energy-saving techniques in the IoT and introduced a new coding scheme for delta compression to efficiently compress temporally correlated data. By reducing the amount of transmitted data, energy consumption is significantly reduced, especially in scenarios where real-time measurements are not required. The developed coding scheme demonstrates energy savings of up to 85% and outperforms other techniques in terms of compression ratio and memory requirements.

In order to achieve energy efficiency in data communication within the IoT, it is essential to adopt a comprehensive approach that encompasses various elements. These elements include data aggregation, the utilization of efficient protocols, the implementation of power management techniques, the integration of edge computing, and the application of intelligent data processing. By employing these tactics, IoT systems have the potential to attain substantial energy conservation, hence prolonging the operational lifespan of devices and facilitating sustainable and scalable IoT deployments. The main contributions exerted for energy-efficient data transfer in IoT that are presented in this section are summarized in Table VI.

III. Taxonomy of IoT DQM Based on Application

This section serves as a foundational component by highlighting the role of IoT big data applications in driving insights, optimizing operations, and delivering value across diverse domains. More particularly, this section systematically presents the most common IoT application areas and illustrates how each is susceptible to distinct security threats that directly impact DQM. As shown in Fig. 5, this section not

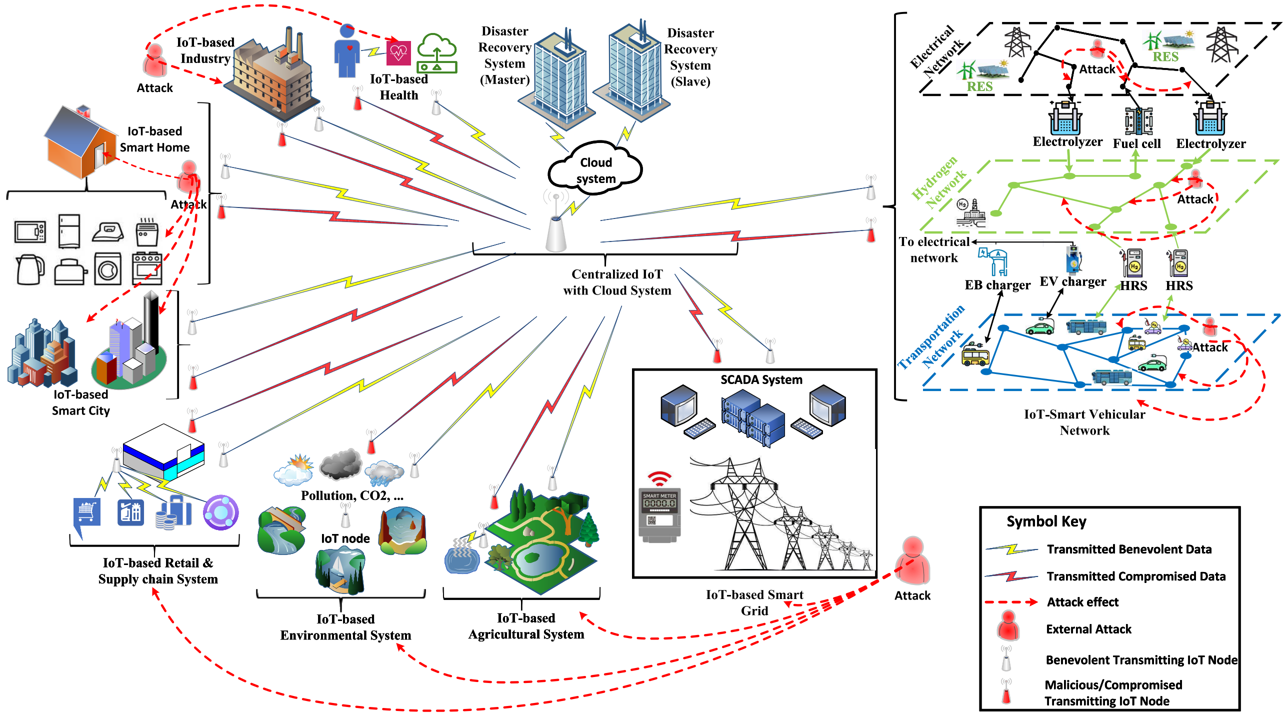

Fig. 5. Common IoT applications along with a centralized IoT system with attack existence based on a synthesis of representative studies from 2016 to 2025.

TABLE VI
TECHNIQUES FOR ENERGY-EFFICIENT DATA TRANSFER IN IOT

Ref	Description
[133]	Presented a cell-free IoT system based on a wireless communication network to enhance reliability and sustainability.
[134]	Proposed AM-DR model in IoT networks to minimize communication costs and energy consumption.
[135]	Proposed DEDaR approach to minimize data transmission, storage, and energy consumption.
[136]	Introduced an energy-efficient IoT communication model based on existing WiFi infrastructure to avoid monthly service charges and reduce power consumption.
[137]	Employed a mixed integer linear programming model to balance reliability and power consumption in IoT applications.
[138]	Developed a coding mechanism for delta compression to efficiently compress temporally correlated data that achieved significant energy savings.

only categorizes these applications but also underscores their relevance to data integrity, accuracy, and usability.

Furthermore, we introduce a taxonomy of IoT DQM in Table VII, which organizes applications based on their primary objectives and the benefits they offer. This taxonomy is further enriched by a discussion of associated challenges and representative use cases, providing a practical and contextual understanding of DQM implications across application domains.

A. Smart Cities

Smart cities involve IoT networks to build an interconnection ecosystem where different urban services and infrastructures are combined and managed in real time [139]. To create more sustainable and effective cities, in [140] and [141], the objective was to solve urban issues such as traffic congestion, energy management, waste management, environmental monitoring, and public safety. Smart cities use IoT technology to improve a range of urban services such as public safety, waste management, smart infrastructure, mobility, utilities, and environmental monitoring [142]. IoT sensors gather and analyze data from buildings, electrical grids, and roadways to manage traffic, optimize performance, and lower energy usage. Real-time monitoring and dynamic adjustments are desired for smart utilities such as electricity and water, while environmental monitoring guarantees improved air management, noise, and water quality [143], [144]. Besides, connected surveillance and emergency response systems enhance public safety, and IoT-enabled monitoring and predictive analysis boost waste management's efficiency while lowering costs and having a positive environmental impact [145], [146].

The degree of smartness in a city directly shapes its data management demands. While early stage smart cities prioritize basic data collection and storage, advanced cities require scalable architectures, real-time analytics, and robust data governance to support integrated systems such as transportation, energy, and healthcare while ensuring low-latency processing and strong data privacy protection [52], [147], [148].

B. Healthcare

IoT-based healthcare describes the combination of intelligent sensors that gather and send health data online [149], [150]. These sensors can be utilized for hospital operations, controlling chronic illnesses, keeping track of patients, and increasing their entire experience [151], [152]. Accordingly, it provides lower costs, enhanced patient outcomes, and better treatment. Adil et al. [153] surveyed several security challenges regarding the wearable devices of patients and focused

TABLE VII
IoT Applications: Target, Key Benefit, Challenges, and Examples

Application	Main Target	Key Benefits	DQM Challenges	Examples	Ref
Smart Cities	Traffic management and monitoring	Less congestion, optimized public transport, Efficient street lighting	Scalability, data security, infrastructure costs	Barcelona, London, and Singapore Smart City & Traffic	[177]
	Waste management	Efficient waste collection, cleaner cities	Data integration, sensor maintenance	Smart waste management in India	[178]
Healthcare	Remote patient monitoring	Improved patient care, reduced hospital visits	Data privacy, compliance with regulations	Telemedicine, wearable E-health devices	[179]
	Smart medical devices	Real-time data, automated diagnostics	Device interoperability, high costs	Smart insulin pumps, connected pacemakers	[180], [181]
Agriculture	Precision farming	Improved crop yield, efficient water use	High setup cost, sensor accuracy	Precision Agriculture	[182]
	Livestock monitoring	Animal health monitoring, optimized feeding systems	Network connectivity, power consumption	Smart collar for cattle	[183]
Industry	Predictive maintenance	Reduced downtime, lower operational costs	Data integration, security concerns	Predix platform	[184]
	Inventory management	Efficient inventory tracking, real-time data insights	Sensor calibration, system interoperability	Siemens IoT solutions	[185]
Environment	Air quality monitoring	Pollution tracking, real-time alerts	Sensor reliability, data storage, and analysis	Air quality egg project	[186]
	Water quality monitoring	Contamination detection, resource management	Network connectivity, device maintenance	Smart water networks	[187]
Smart Home	Energy management	Reduced energy consumption, cost savings	Device compatibility, privacy issues	Google Nest Thermostat	[188]
	Home security	Real-time monitoring, improved safety	Network vulnerability, data breaches	Ring Video Doorbell	[189]
Retail and Supply Chain	Inventory management	Real-time stock updates, optimized supply chain	System integration, data security	Amazon smart warehouse	[190]
	Customer behavior analysis	Personalized marketing, improved customer experience	Data privacy, data processing speed	IoT-enabled smart shelves	[191]

on the observation, processing, storage, and evaluation of data. Relying on smartwatches and smartphones has been studied to monitor the lifestyle of patients as a real-time application [154]. That work has employed a fast protocol called Fast Healthcare Interoperability Resources. DL and lightweight authentication research efforts have also been extended to biomedical IoT-based healthcare to enhance seizure recognition and e-healthcare, as studied in [155] and [156].

C. Agriculture

Employing IoT in agriculture represents a valuable step toward modernizing traditional farming practices [157], [158]. On the one hand, IoT sensors can provide real-time data on critical variables such as soil moisture, temperature, humidity, and nutrient levels. This allows farmers to make suitable decisions, optimize irrigation schedules, and prevent water wastage and crop damage due to underwatering or overwatering [159], [160].

IoT can also play a vital role in monitoring crop health, in which sensors can detect disease indications, pest infestations, or nutrient deficiencies early on [161]. In this regard, employing data analytics can assist farmers in addressing these issues promptly, reducing the reliance on chemical treatments, and enhancing the overall health of the crops [162]. Furthermore, IoT systems improve yield predictions by continuously gathering and analyzing data on plant growth, environmental conditions, and past harvest trends. Consequently, IoT system-based farming can contribute to enhanced harvesting for better future market demands, increasing profitability while promoting more sustainable farming practices [163].

D. Industry

IoT is considered the key player in the industrial revolution. IoT has contributed to developing the manufacturing and production processes by enhancing efficiency, reducing costs, and improving overall productivity [164], [165]. In other words, IoT devices such as sensors, actuators, and intelligent controllers enable real-time monitoring and control of machinery, equipment, and production lines [166]. Accordingly, these systems can detect inefficiencies, reduce downtime, and optimize energy consumption. Industrial automation and real-time monitoring are among the key benefits of IoT, especially from the point of view of predictive maintenance [167]. IoT sensors can continuously monitor many essential aspects of the industrial process, such as the health of machines. It can track vital parameters, e.g., vibration, temperature, wear, and waste [168], [169], [170]. Therefore, analyzing such data can early detect potential issues, which is considered a time constraint for smooth maintenance and reducing unexpected breakdowns, leading to significant cost savings [171]. This approach minimizes downtime and extends the lifespan of equipment, leading to significant cost savings. More concretely, IoT systems can operate the manufacturing process more wisely by integrating data from various stages of production [172]. This can achieve better coordination between different processes, keep production lines synchronized, and operate at optimal efficiency [173]. The data collected can be used for real-time decision-making, quality control, and process optimization, leading to higher output quality and faster response to market demands [174].

Countries are adopting diverse strategies to address IIoT data challenges. Germany emphasizes edge computing under Industry 4.0 for low-latency processing, the U.S. focuses on blockchain-based security for data integrity, and China leverages AI-driven anomaly detection for predictive maintenance. These approaches reflect varied global efforts to enhance IIoT DQ, security, and responsiveness [175], [176].

E. Environmental Monitoring

IoT plays a vital role in environmental monitoring by providing real-time data to track and manage different environmental conditions, such as line of sight, nonline of sight, indoor, outdoor, underground, and underwater, more efficiently [192], [193], [194], [195]. IoT sensors can be deployed in various environments—forests, oceans, and cities—to monitor air and water quality, detect pollution, track wildlife, and measure parameters such as temperature, humidity, and CO_2 levels [186], [188].

For air quality, IoT sensors can continuously monitor pollutants such as carbon dioxide and particulate matter, enabling authorities to identify pollution sources and take immediate action. Similarly, IoT devices in water bodies can measure factors like pH, salinity, and chemical contaminants, helping detect pollution events early and protecting ecosystems.

IoT-driven environmental monitoring systems can also provide crucial data for disaster prevention. For instance, forest sensors can detect early signs of wildfires by measuring heat and humidity, allowing for rapid response. In addition, these systems aid in tracking wildlife movements, studying migration patterns, and ensuring species conservation.

F. Smart Homes

Employing IoT at home has facilitated appliances' remote use and enhanced the interconnection between them. This smart home automation improved the daily lifestyle and assisted disability people to live normally [193]. Smart homes rely on intelligent sensors such as thermostats and lighting systems, which can be controlled remotely or configured to operate automatically based on user needs. For instance, smart thermostats can automatically schedule, adjust heating or cooling, and improve energy efficiency [196].

An important part of IoT-based smart homes is the security front. Accordingly, some desired features are engaged, such as smart locks, surveillance cameras, and motion sensors. These features enable homeowners to remotely monitor and secure their property as a real-time application using, for example, mobile apps [197]. Moreover, IoT enables real-time monitoring and optimizing energy usage [198]. This can lead to cost savings and a reduced environmental footprint, as lighting or heating only runs when needed [199], [200].

G. Retail and Supply Chain

The adoption of IoT in retail and supply chain management has improved the efficiency, visibility, and customer experience [201]. For example, RFID tags, smart sensors, and connected vehicles help track inventory as a real-time application, leading to reducing the risk of stock shortages or overstocking [190], [191]. Such a process can assist the entire supply chain in enhancing data-driven decisions, improving forecasting, reducing waste, and streamlining logistics [202]. In retail, IoT enhances customer experiences through smart shelves, automated checkouts, and personalized recommendations based on consumer behavior. Smart shelves can detect when stock levels are low and automatically trigger reorders, while connected payment systems offer faster contactless transactions and reduce the risk of theft [190], [191].

H. Smart Grids

DQM in IoT-based SGs plays a critical role in ensuring efficient and reliable operation [203], [204]. Indeed, SG is a real-time application that relies on many IoT devices, such as sensors, smart meters, and control systems that generate massive volumes of data [205]. Accordingly, the observed data must be collected, processed, and analyzed efficiently to optimize energy distribution, detect faults, and enhance overall grid performance [206], [207], [208], [209]. Effective DQM does not only rely on the storage and information handling but also on the use of advanced analytics. For example, ML is used to predict energy demand patterns and prevent grid outages [210], [211]. It also ensures essential aspects such as data integrity, security, and privacy to address potential vulnerabilities in the grids' communication links. Besides, efficient DQM enables SGs to become more flexible, scalable, and resilient to changing energy demands.

IV. Roles of Modern Technology in IoT DQM Using: Processing, Analysis, and Security

This section highlights the fast-booming technologies that contributed to effective DQM in IoT networks relying on data processing, analysis, and security. In this regard, we focus on edge computing, ML, and blockchain.

A. Edge Computing for IoT Data Processing

The concept of edge computing pertains to a decentralized computing model wherein the processing and storage of data occur in analogy to the network edge, near the origin of data generation [212], [213]. The general scheme of a fog computing architecture is represented in Fig. 6. In contrast to conventional cloud computing, which is dependent on centralized data centers, edge computing entails the deployment of computer resources and services in closer proximity to the devices and sensors located at the network edge [214]. The closeness between components facilitates expedited data processing, decreased latency, heightened scalability, augmented privacy, and instantaneous decision-making [215], [216]. Edge computing enables the optimization of network bandwidth and facilitates the execution of time-sensitive applications, such as IoT devices, autonomous systems, and real-time analytics, by relocating computational workloads and data storage in proximity to the network edge [217], [218]. In general, edge computing facilitates the ability of companies to locally process and analyze data at the edge devices, hence mitigating the necessity for constant data transmission to faraway cloud infrastructures [219], [220], [221].

Edge computing offers several advantages for IoT data management, including reduced latency, improved performance, bandwidth optimization, enhanced reliability, increased

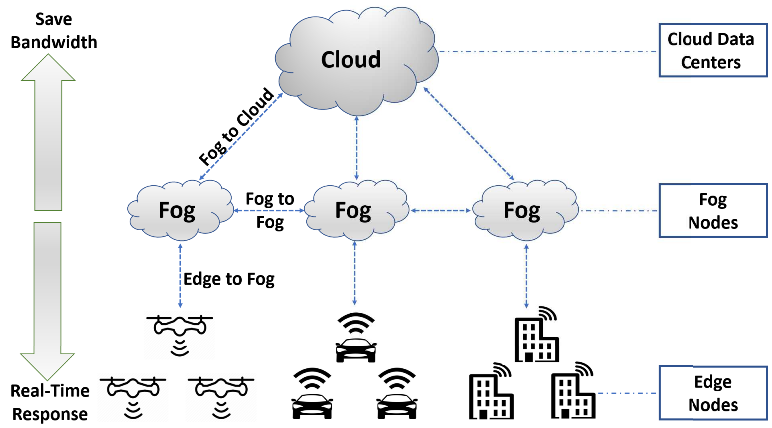

Fig. 6. Fog computing general scheme.

TABLE VIII
MAIN BENEFICIAL PARAMETERS OF FOG COMPUTING IN IoT

Parameter	Description	Parameter effect	Ref
Reduced Latency	Edge computing allows for faster data processing and analysis by bringing computation closer to the source of data generation, reducing the time it takes for data to travel to a centralized cloud or data center.	Low latency, closer to the edge devices than cloud	[222]
Improved Performance	By processing data at the edge, edge computing can provide real-time decision-making capabilities, enabling faster response times and improved overall performance of IoT applications.	Enhances performance by offloading tasks from the cloud	[223]
Bandwidth Optimization	Edge computing helps optimize network bandwidth by reducing the amount of data that needs to be transmitted to the cloud, alleviating network congestion, and reducing bandwidth requirements.	Reduces bandwidth usage by filtering data locally	[224]
Enhanced Reliability	With edge computing, local processing, and decision-making can still occur even when the network connection to the cloud is disrupted, improving the reliability and resilience of IoT applications.	High, as fog nodes can handle processing even when disconnected from the cloud	[225]
Increased Privacy & Security	Edge computing allows data to be processed and analyzed locally, reducing the need to transmit sensitive data to the cloud, thereby enhancing privacy and security.	Improves data privacy by processing sensitive data locally	[226]
Scalability	Edge computing enables distributed computing resources, allowing for scalable and flexible deployment of IoT applications, especially in remote or resource-constrained environments.	Moderately scalable, supports adding more fog nodes between cloud and edge	[227]
Cost-effectiveness	By reducing the need for extensive data transmission and centralized cloud resources, edge computing can help lower costs associated with data management and processing for IoT applications	More cost-effective than pure cloud solutions due to reduced data transfer	[228]

privacy and security, scalability, and cost-effectiveness as discussed in Table VIII.

The constraints associated with edge computing in the context of IoT data management encompass certain restrictions in theoretical analysis stemming from the inherent unpredictability of the Internet, divergent routing practices employed by Internet and cloud providers, as well as the comparatively restricted resource capacities of edge devices about cloud servers [229], [230]. Furthermore, it is worth noting that the current data error-finding mechanisms in blockchain-enabled edge computing schemes exhibit suboptimal efficiency. Additionally, compared to cloud-based solutions, the scalability and responsiveness of edge applications remain constrained. It is imperative to take into account the limitations in resources and the expenses related to energy consumption in the context of IoT devices. Additionally, there is a necessity to prioritize the effective allocation and administration of resources at edge devices to guarantee a satisfactory level of service for IoT applications. In summary, edge computing has notable benefits for the administration of IoT data. These advantages encompass diminished latency, enhanced performance, optimized bandwidth utilization, augmented reliability, heightened privacy and security measures, scalability, and cost efficiency.

B. ML for IoT Data Analysis

Nowadays, the use of ML has not become an option. ML has enhanced classical programming in many fields [12], [29], [30], [231], [232]. Fig. 7 shows the difference between the main component of classical programming and ML. Indeed,

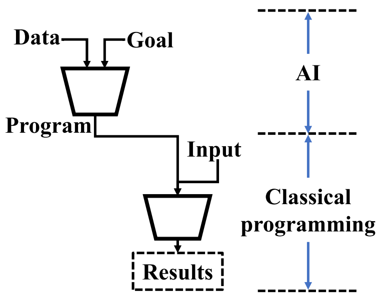

Fig. 7. Classic programming versus ML.

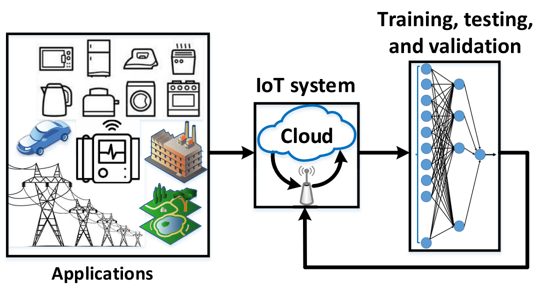

Fig. 8. Architecture of ML role for IoT data analysis.

ML has been extensively employed for examining the large quantities of data produced by IoT devices [233], [234]. More particularly, SGs, smart homes, smart cities, and smart vehicles are among the IoT-based applications in which the DQ can be preserved using modern data prediction and analysis such as ML as illustrated in Fig. 8. With the increasing proliferation of interconnected physical devices, there arises a corresponding demand for effective approaches to extract meaningful insights from the data they provide [235]. ML algorithms present a viable alternative by offering automated and intelligent methodologies to effectively process and derive meaningful insights from IoT data [236]. By utilizing these algorithms, it becomes possible to identify trends, detect abnormalities, and make predictions [50]. This capability empowers enterprises to optimize their operations, better their decision-making processes, and improve the overall functionality of IoT devices [237]. Table IX categorizes suitable samples of the efforts exerted for IoT data management using ML models taking into consideration the application type, data type, challenge, employed ML model, cloud dependency, and performance evaluation metrics.

C. Security Aspects in IoT Data Management

Security is a vital part of IoT data management that must be addressed to ensure the protection and integrity of sensitive data gathered and processed by IoT devices with many applications, such as smart homes, industry, autonomous vehicles, and SGs. There are several essential issues for ensuring robust security as the number of connected devices grows and data travels between them and cloud-based platforms.

Organizations may improve the security of IoT data, retain user confidence, and reduce the risks associated with unauthorized access, data breaches, and privacy violations by addressing these security factors. To overcome these factors, a multilayered approach to security is required, requiring a combination of safe hardware design, robust software development techniques, effective security policies, and continual monitoring and reaction mechanisms. This section summarizes the security aspects into three main categories, i.e., data protection and privacy, system security and Infrastructure, and threat management and response as discussed in Sections II-H1–II-H3. Table X presents a comprehensive taxonomy and summary for securing and managing IoT data concerning the security aspects and a brief description for each, the role of each security method in IoT data management, and the adopted solutions.

1) Data Protection and Privacy: For attaining efficient IoT data management via data protection and privacy, five security aspects should be guaranteed. First, end-to-end security can attain comprehensive protection, mitigate the risk of data breaches, and ensure trusted compliance [274], [275], [276]. Second, data sovereignty can guarantee regulatory compliance of data, ensure user trust and control, and avoid legal complexities [277], [278], [279]. Third, efficient data privacy achieves user confidentiality, minimizes the risk of data misuse, and meets privacy laws [280], [281], [282], [283], [284]. Fourth, data description is responsible for data confidentiality, data tampering prevention, and ensuring safe storage and transmission [47], [285], [286], [287]. Fifth, a secured communication protocol preserves the integrity of data transfers, achieves authentication and authorization, and serves real-time threat detection [288], [289], [290], [291], [292].

2) System Security and Infrastructure: In this category, six security aspects should be guaranteed. First, secure cloud infrastructure is beneficial for data integrity and availability, scalable security, disaster recovery and data backup, and regulatory compliance [27], [293], [294], [295]. Second, authentication and access control are desirable for the prevention of unauthorized access, granular permission control, multifactor authentication, and auditable access logs [296], [297], [298]. Third, secure boot and hardware security are employed for trusted device start-up, protection against physical attacks, and improving device integrity [299], [300], [301], [302]. Fourth, network segmentation is suitable for minimizing attack surface effect, better traffic monitoring, attack containment, and enforcing security policy [303], [304], [305]. Fifth, a device management upgrade is required for remote monitoring and updates, and secure firmware updates [300], [306], [307], [308]. Sixth, key management is demanded for establishing efficient secured keys and enhancing operational efficiency [309], [310].

3) Threat Management and Response: In this category, five security aspects have to be maintained. First, IDPSs are used to enhance early threat identification, mitigate real-time attacks, improve system visibility, and compliance support [53], [311], [312], [313], [314]. Second, regular security audits and monitoring can contribute to managing proactive vulnerability, continuous improvement, early detection of security gaps, and compliance assurance [315], [316], [317], [318]. Third, resilience and recovery plans are involved in minimizing downtime, data preservation, business continuity, and incident response [319], [320], [321]. Fourth, supply chain security is acquainted with protection against third-party risks, the integrity of hardware and software, and secured delivery channels [322], [323], [324]. Fifth, threat detection is utilized for advanced malware, detection behavioral analysis, rapid incident response, and protection from zero-day exploits [325], [326], [327], [328], [329], [330].

TABLE IX
ML FOR IoT DATA ANALYSIS

Application	Target	Data type	ML model	Cloud dependency	Accuracy Evaluation	Ref
Smart cities	Traffic classification	20 days of public data records	several classical ML classifiers, ANN, CNN, and LSTM	N/A	Overall accuracy, F1-score, precision, and recall	[238]
	Accident detection	RGB frames based on optical flow information	Hybrid model via CNN and LSTM	N/A	Mean average precision	[239]
	Data transmission through secured medium	Dataset source [240]	FL	✓	F-scores, accuracy, recall, and precision	[241]
	Handling emergencies via transaction prioritization	Synthetic dataset	LightGBM enhanced by Blockchain	-	Overall accuracy, trustworthiness score, and loss	[242]
	Managing fault tolerance & detecting malicious nodes	Synthetic dataset	Supervised ML	✓	Benevolent data delivered rate, transmission cost, communication overhead, energy consumption	[243]
Healthcare	Patients private information protection	COVID19 chest X-ray, STL10, MNIST, and CIFAR10	FL & CNN	✓	Accuracy, MAE	[244]
	Patients data privacy	Datasets from ASLDVS, N-MNIST, N-CARS, NCaltech101, CIFAR10-DVS [245], [246]	Adversarial ML mechanism	✓	Accuracy	[247]
	Detecting heart disease risk	Source [248]	Five classical ML models	✓	F-scores, accuracy, recall, precision, and AUC	[249]
	Mitigating cyber attacks in Industry 5.0-based healthcare	NSL-KDD and KDD CUP 99 datasets [250], [251]		✓	Computational complexity, F-scores, accuracy, recall, and precision	[252]
Agriculture	Fault diagnosis in farming application	Simulated via NS 2.35	SVM and reinforcement learning	N/A	Accuracy, energy consumption, F1 score, delay, and lifetime	[253]
	Assessment of end-to-end DQ	Dataset captured by weather stations in the UK from 2015 to 2019	ML model impeded in big data cycle	✓	RMSE and MAE	[254]
	Predicting crop pest	temperature, humidity, and rainfall datasets collected by CC2538-2592 SoC microcontroller with IEEE 802.15.4 standard	Fuzzy logic	✓	Cauchy membership function	[255]
	Detecting soil fertility level	Local dataset captured by fertility sensors for soil characteristics	Light gradient boosting (LGB)	✓	MSR, RMSE, MAE	[256]
Industry	Data analytics for Industry 5.0-based IIoT systems	CICIDS2017 and IoTID20 datasets	Ensemble learning	✓	F-scores, accuracy, recall, precision, latency, and throughput	[257]
	Prediction of product quality	Observed mineral purification by soft sensor	Deep factorization machine	N/A	MSE, RMSE	[258]
	Monitoring of industrial dynamic process	Synthetic data	Stacked autoencoder along with spatiotemporal neighborhood	N/A	Computational time, R2, and RMSE	[259]
	Monitoring of dairy products packaging	Logs of packag-ing alarm [260]	Transformer	N/A	precision, recall, F1-score, and loss function	[261]
Environment	Discovery the events of indoor thermal comfort	Observational data [262]	Causal ML	N/A	Precision, Recall, F1-score	[263]
	Detecting pollutant in aquatic environments	Synthetic data	Multilayer perceptron	✓	Accuracy and MSE	[264]
	Monitoring Water in a Lentic Ecosystem	Real observed data from a lake	Multi-ML models	N/A	Heat map pearson correlation coefficient	[265]

TABLE IX
(Continued.) ML FOR IOT DATA ANALYSIS

Application	Target	Data type	ML model	Cloud dependency	Accuracy Evaluation	Ref
Smart home	Detecting the intrusion in MQTT protocol	MQTT-IoT-IDS2020	Multi-ML classifiers	N/A	Overall accuracy and F1-score	[266]
	Protecting the technology of fiber-to-the-home	FTTH complaint log driven from a dataset from Philippines SKY cable company [267]	KNN	-	Accuracy, F1-score, precision, and recall	[268]
	Detecting malicious activities	NSL-KDD dataset	LSTM along with blockchain	✓	Accuracy, F1-score, precision, recall, confusion matrix, and MCC	[269]
	IDS	DS2OS	LGB, logistic regression, extreme GB, random Forest	N/A	Accuracy, MSE, RMSE, MAE, confusion matrix, and ROC	[270]
Retail and Supply Chain	COVID-19 vaccine supply chain management	Observed data from warehousing, industrial logistics, and transportation systems	GRU and blockchain	-	Accuracy and RMSE	[271]
	Securing end-to-end food supply chain	Synthetic data	TinyML	✓	Accuracy and F1-score	[272]
	Food traceability system including e-commerce transactions		Watson ML and Fuzzy logic	✓	Transition time	[273]

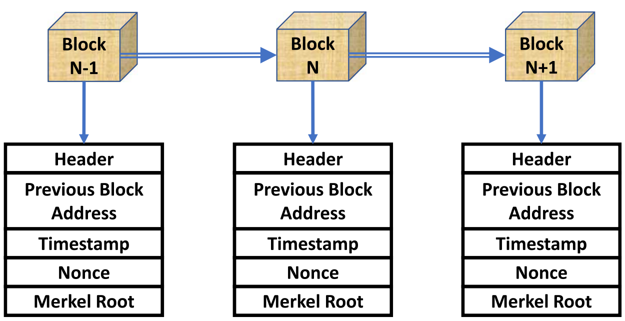

Fig. 9. Blockchain general architecture.

D. Blockchain for IoT Data Access

Blockchain technology has received considerable attention due to its promise to improve security, transparency, and trust in a variety of industries [331]. When applied to IoT data access, blockchain can provide various advantages and handle issues such as data integrity, authentication, and decentralized control. The architecture of blockchain is shown in Fig. 9. Here is an outline of how blockchain technology can be used to access IoT data [332], [333].

Interestingly, by leveraging blockchain, IoT systems can ensure tamper-proof logging of sensor data, thereby enhancing the trustworthiness and traceability of recorded events [334]. Smart contracts can further automate secure access control policies, allowing only authorized entities to retrieve or modify data in a decentralized fashion. This removes reliance on a centralized authority, which is often a point of failure and vulnerability in traditional IoT architectures. Furthermore, blockchain aids in auditability and accountability in multistakeholder environments such as smart cities, supply chains, and healthcare systems, where data ownership and provenance are critical [335]. However, integrating blockchain into IoT systems also raises several challenges. The scalability of blockchain remains a concern due to the limited processing capabilities of edge devices and the high computational cost of consensus algorithms [336]. Latency introduced by blockchain operations may conflict with the real-time requirements of some IoT applications [337]. Additionally, the storage overhead of maintaining a distributed ledger across constrained IoT nodes is a significant hurdle. Therefore, lightweight blockchain frameworks, off-chain storage strategies, and hybrid consensus models are being explored to address these limitations [338]. Besides, some open concerns must be addressed when using blockchain technology for IoT data access, as extensively categorized in Table XI. These include privacy-preserving mechanisms, interoperability with existing IoT protocols, energy efficiency, and regulatory compliance.

V. RESEARCH TRENDS, OPEN ISSUES, CHALLENGES, AND FUTURE DIRECTIONS

The rapid evolution of IoT data management is expected to progress along several significant research trajectories. This section addresses the main research trends, open issues, challenges, and future directions shaping the future landscape of IoT data ecosystems. The identified research trends are categorized into ten key domains: IoT and smart cities, ML/AI in IoT, security and privacy, big data/cloud/edge/fog computing, healthcare IoT, energy/SG, agriculture/environment monitoring, data management/quality, Industry 4.0/smart manufacturing, and other (general concepts). These trends are analyzed from both annual (i.e., from 2016 to mid-2025) and thematic perspectives, as illustrated in Figs. 10 and 11, respectively.

TABLE X
SECURITY IN IoT DATA MANAGEMENT

Security Category	Aspect	Brief Description	Security Aspect Role in IoT Data Management	Common Solutions/Technologies	Ref Sample
Data Protection and Privacy	**End-to-End Security**	Establishes security measures across all stages of data handling, from collection to storage.	Ensures that data is protected throughout its lifecycle, including transmission and storage.	End-to-End Encryption (E2EE), full-stack security, zero trust architectures	[274]–[276]
	Data Sovereignty	Ensures IoT data is stored and processed in compliance with local laws on data storage and privacy.	Important for regions with strict data sovereignty laws, ensuring legal compliance.	Data localization services, sovereign cloud services, regional data centers	[277]–[279]
	Data Privacy	Ensures personal or sensitive data collected by IoT devices is handled according to privacy laws and user consent.	Vital for compliance with privacy regulations (e.g., GDPR), ensuring that user data is not exploited or leaked.	Data anonymization, GDPR compliance tools, privacy by design approaches	[280]–[284]
	Data Encryption	Protects data by encoding it during transmission or storage, so it is unreadable without the correct key.	Essential for securing sensitive data both in transit and at rest, safeguarding it from interception or theft.	AES, RSA, TLS/SSL, E2EE	[47], [285]–[287]
	Secure Communication Protocol	Protocols that ensure data transmission between IoT devices is secure from interception and manipulation.	Provides integrity and confidentiality of communications between IoT devices and servers.	MQTT with SSL/TLS, HTTPS, DTLS, CoAP, IPsec	[288]–[292]
System Security and Infrastructure	**Secure Cloud Infrastructure**	Provides a secure environment for IoT data storage and processing on cloud platforms, protecting against breaches and attacks.	Necessary for ensuring the confidentiality, integrity, and availability of data in cloud services used by IoT systems.	Secure cloud storage (AWS IoT, Azure IoT Hub), encryption, identity access management (IAM) for cloud infrastructure, cloud access security brokers	[27], [293]–[295]
	Authentication and Access Control	Ensures only authorized users and devices can access the system or data.	Critical for preventing unauthorized access to sensitive IoT data and controlling resource usage.	Multi-factor authentication, OAuth, role-based access control, IAM	[296]–[298]
	Secure Boot and Hardware Security	Ensure IoT devices boot only with trusted and verified software to prevent malicious code execution.	Essential for protecting devices from hardware-level attacks and ensuring software integrity.	Trusted platform module, secure boot, ARM TrustZone, Intel SGX	[299]–[302]
	Network Segmentation	Divides networks into smaller segments to limit the spread of breaches or attacks.	Reduces the attack surface and limits the lateral movement of threats within IoT networks.	Virtual LAN, software-defined networking, micro-segmentation	[303]–[305]
	Device Management and Upgrade	Involves regular firmware updates and patch management to fix vulnerabilities in IoT devices.	Critical to maintaining device security over time, especially as new threats emerge.	Over-the-Air updates, device configuration management, centralized firmware management Systems	[300], [306]–[308]
	Key Management	Secure management of encryption keys for authentication, communication, and data protection.	Ensures that encryption and authentication processes remain secure and uncompromised.	Public key infrastructure, HSM, secure key distribution and rotation protocols	[309], [310]
Management Threat and Response	**Intrusion Detection and Prevention Systems (IDPS)**	Monitor the network for malicious activity or violations of security policies, and take action to prevent threats.	Important for early detection of attacks, ensuring real-time response to malicious intrusions in IoT networks.	IDPS systems, anomaly detection algorithms, security information, and event management)	[53], [311]–[314]
	Regular Security Audits and Monitoring	Involves continuous security assessments and audits to ensure the IoT environment remains secure and compliant.	Critical for identifying vulnerabilities and ensuring security measures remain effective as the IoT environment evolves.	Vulnerability scanning, penetration testing, security operations centers, continuous monitoring tools	[315]–[318]
	Resilience and Recovery Plans	Plans for ensuring quick recovery from cyberattacks or failures, focusing on business continuity.	Critical for reducing downtime and impact during security breaches or system failures.	Disaster recovery plans, backup solutions, incident response frameworks	[319]–[321]
	Supply Chain Security	Protects against vulnerabilities introduced during the manufacturing or deployment of IoT devices.	Essential to avoid compromised hardware or software entering IoT systems through third-party components.	Vendor security audits, secure supply chain standards (NIST, ISO/IEC 27001), blockchain for supply chain integrity	[322]–[324]
	Threat Detection	Uses AI and ML to detect anomalies and security threats in IoT networks in real-time.	Increases the detection accuracy for new or unknown attack patterns.	AI-based IDS, ML-based Anomaly detection, behavioral analytics	[325]–[330]

TABLE XI
COMPARATIVE ANALYSIS OF BLOCKCHAIN FOR IOT DATA ACCESS

Feature	Brief Description	Advantages	Challenges	Application	Ref
Decentralized Data Access Control	Eliminates centralized authority, allowing devices to directly control access to their data using smart contracts and consensus algorithms.	• Reduces reliance on centralized servers • Lowers single points of failure • Ensures autonomous device operation	• Scalability issues due to increased data traffic • Throughput and latency concerns with real-time access • Energy inefficiency from consensus mechanisms • Governance challenges related to decentralized access rights	• Smart homes • Smart cities	[334], [339], [340]
Data Integrity and Auditability	Ensures data remains unaltered by recording all transactions in immutable ledgers, providing an auditable trail.	• Guarantees tamper-proof records • Boosts trust • Enables regulatory compliance	• High storage costs for large-scale IoT data • Throughput challenges from large datasets • Privacy concerns during auditing • Energy consumption in maintaining immutable ledgers	• Healthcare applications • Patient data management	[332], [341], [342]
Enhanced Security and Authentication	Enhances security via cryptographic techniques and consensus, preventing unauthorized access or tampering.	• Provides robust protection against cyberattacks • Ensures secure data sharing • Improves user trust in IoT systems	• Latency from consensus verification in large-scale IoT • Interoperability issues with existing security frameworks • Energy inefficiency in cryptographic processing • Regulatory concerns in multi-region authentication systems	• Supply chain management • Secure, authenticated tracking of goods	[343]–[345]
Data Monetization and Ownership	Enables IoT devices and data owners to monetize data through secure, traceable exchanges while maintaining ownership rights.	• Encourages economic incentives for data sharing • Preserves ownership rights • Facilitates new business models	• Lack of clear standards for data monetization • Privacy risks in data ownership • Complex governance structures for fair compensation • High cost of implementing blockchain-based monetization systems	• Energy markets • Smart devices trading surplus energy	[333], [346]
Data Sharing and Collaboration	Facilitates secure and transparent data exchange between IoT devices or entities, allowing seamless collaboration without intermediaries.	• Promotes transparent peer-to-peer interactions • Enhances collaboration in IoT ecosystems • Reduces data silos	• Interoperability issues with diverse blockchain platforms • Scalability challenges in large networks • Privacy concerns in data sharing • Governance and standardization hurdles for cross-border collaboration	• Collaborative IoT applications • Agriculture and industrial automation	[332], [344]

Following this analysis, we shed light on the prevailing open issues, unresolved challenges, and promising directions for future research in the field as follows.

A. AI-Powered Data Insights

The integration of AI and IoT data management represents dependent technologies in DQM. ML algorithms and DL models are expected to play a critical role in deriving complex insights from massive amounts of IoT data. These algorithms are capable of detecting subtle patterns, correlations, and anomalies that standard analytical methods may miss. As AI models grow, they will provide decision-makers with more precise, context-aware intelligence. Furthermore, advances in edge AI, which perform ML computations directly on IoT devices, promise to bring real-time, localized decision-making to the forefront.

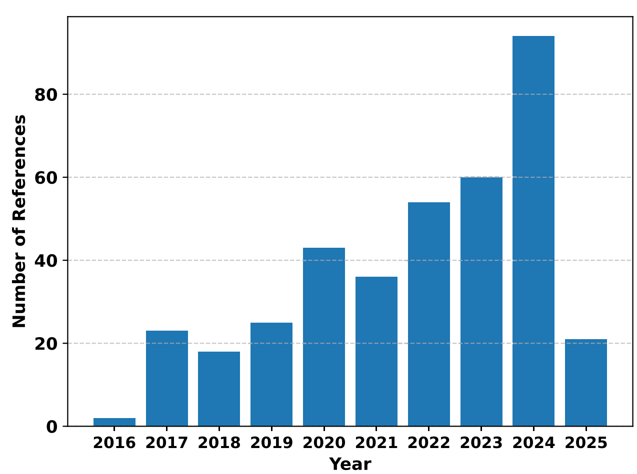

Fig. 10. Main annual focus of IoT DQM research trends.

B. Maturity of Edge Computing

Edge computing's development marks a fundamental change in IoT data processing. Edge devices are becoming more sophisticated in terms of computational capability. Because of this progress, they can now do complicated

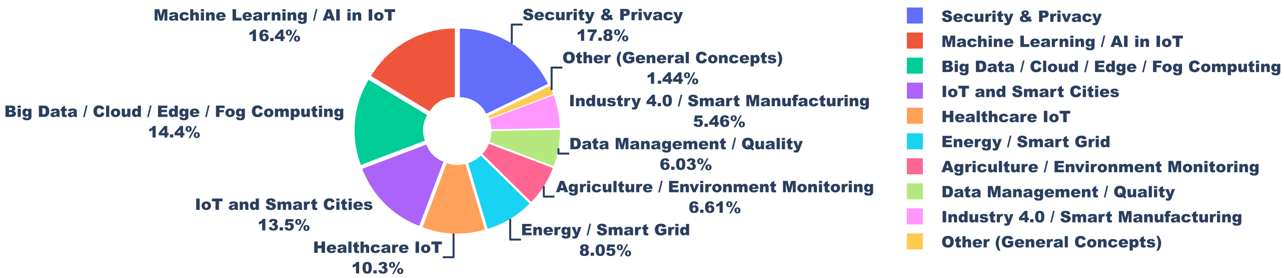

Fig. 11. Thematic perspectives of IoT DQM key domains.

computations locally, minimizing the requirement for continual data transmission to centralized cloud servers. As a result, latency is significantly decreased, allowing for faster reaction times for crucial applications. This trend not only improves real-time decision-making but also reduces the load on centralized cloud infrastructure, resulting in more efficient and responsive IoT installations.

C. Interoperability and Standardization

The seamless interoperability of various IoT devices and platforms is critical for maximizing the potential of IoT ecosystems. It is critical to make efforts to standardize communication protocols and data formats. Not only does standardization allow for simple integration, but it also encourages a collaborative atmosphere in which developers can construct cross-compatible apps. As the IoT ecosystem grows, so will the demand for interoperability, necessitating continued work in standardization and compatibility.

D. Security in the Age of Pervasive Connectivity

The growth of linked devices increases the attack surface, making security a top priority. To protect IoT systems from emerging cybersecurity threats, strong safeguards are required. This includes the implementation of advanced encryption algorithms, decentralized identity management, and intrusion detection systems. Furthermore, the incorporation of secure boot processes and hardware-based security features will be crucial in strengthening the security posture of IoT installations, assuring data integrity and confidentiality.

E. Scalability and Resource Optimization

As the number of connected devices grows quickly, guaranteeing scalability becomes critical. It is a huge problem to design systems that can smoothly handle an increasing number of devices while improving resource efficiency. Continuous research and development in areas such as efficient hardware design, lightweight communication protocols, and distributed computer architectures are required. Innovations in these areas will be critical to allowing IoT implementations to scale smoothly and sustainably.

F. Ethical Considerations and Data Governance

As IoT applications reach more elements of daily life, ethical concerns around data protection, permission, and ownership become more important. It is critical to establish clear data governance rules and adhere to privacy regulations. Strong consent systems, transparent data usage regulations, and tools for individuals to assert control over their own data are all part of this. It is critical to strike a balance between innovation and the protection of individual rights in order to ensure the responsible and ethical use of IoT technologies.

G. Environmental Sustainability

The environmental impact of extensive IoT adoption is becoming increasingly concerning. Efforts must be made to reduce energy use and electronic waste generation. This involves work on energy-efficient hardware design, low-power communication technologies, and environmentally friendly production techniques. Furthermore, developing standards for end-of-life device management and recycling will be critical in reducing the environmental impact of IoT installations.

Navigating these future directions and overcoming the obstacles associated with them will be critical in realizing the full potential of IoT data management. As the IoT landscape evolves, interdisciplinary collaboration and new solutions will be critical in designing an efficient and secure linked world.

VI. Conclusion

The rapid evolution of IoT data management demonstrates the dynamic interaction between technology and the expectations of a connected world. The advances highlighted in this work demonstrate the extraordinary progress made in exploiting the potential of IoT data for a wide range of applications. This tutorial survey has provided a detailed exploration of the evolving landscape of IoT data handling, with a specific focus on maintaining and enhancing DQM. By examining key elements such as secure communication protocols, edge computing, synthetic data usage, and ML for anomaly detection, we have highlighted both the current capabilities and limitations of IoT ecosystems. Emphasis on data access, data analysis, energy-efficient methods, and security measures, including categories such as blockchain, ML, data protection and privacy, system security and infrastructure, and management threat and response, demonstrates the need for a balanced approach to data integrity and system performance. As IoT continues to grow, addressing the open challenges and capitalizing on emerging trends will be crucial for developing more robust, efficient, and secure data management practices. This article aims to serve as a foundational guide for those engaged in the ongoing development and implementation of IoT technologies.

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